EXPERIMENT NO. 12 – LEXICOGRAPHICALLY SMALLEST STRING BY SWAPPING TWO CHARACTERS USING RAPTOR

AIM

To draw and validate a RAPTOR flowchart that swaps all occurrences of two selected characters at most once to obtain the lexicographically smallest string.

PROCEDURE

1. Open RAPTOR and create a new flowchart.
2. Insert Input symbols to read the string str and the two characters ch1 and ch2.
3. Initialize i ← 1, calculate n ← Length_Of(str), and initialize result ← "".
4. Insert a Loop and use Selection symbols to compare str[i] with ch1 and ch2.
5. If str[i] matches ch1, append ch2 to result. If it matches ch2, append ch1. Otherwise, append the original character str[i].
6. After the branches join, increment i ← i + 1 and continue until i > n.
7. After the loop, display result using an Output symbol.
8. Run the flowchart with the sample input and verify the modified string.

OUTPUT

Input: str = ccad, ch1 = c, ch2 = a

Output: Modified string = aacd

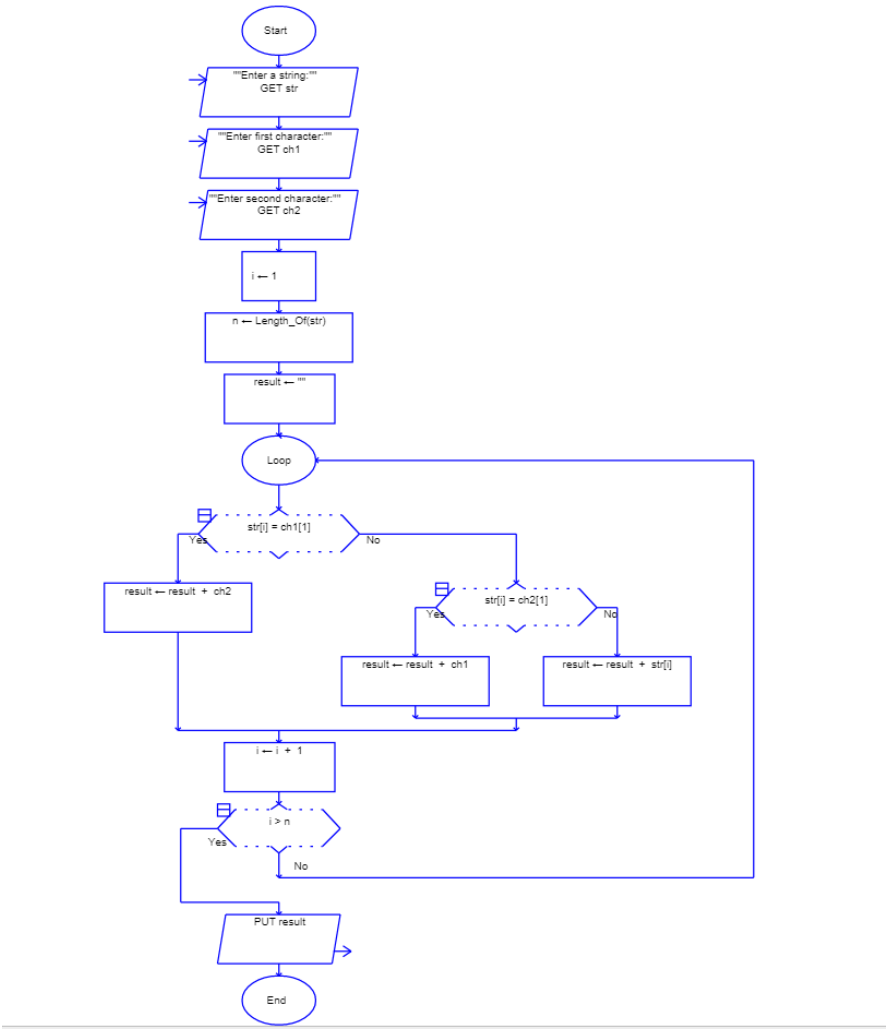


Figure 1 – RAPTOR flowchart

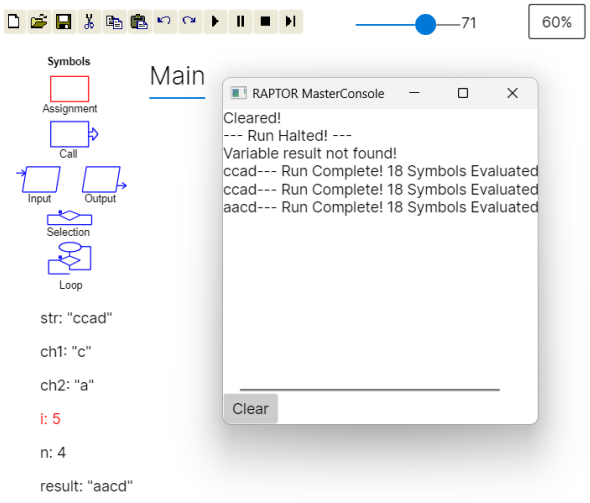


Figure 2 – RAPTOR execution output

RESULT

Thus, the flowchart to obtain the lexicographically smallest string by swapping all occurrences of the selected two characters at most once was drawn and validated successfully using RAPTOR.